

SUNIN BAEK

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EDUCATION

Korea University, Seoul M.S. in Mechanical Engineering	2025 - Present
Korea University, Seoul B.S. in Mechanical Engineering B.S. in Medical Convergence Engineering	2018 - 2025

EXPERIENCE

Movement Intelligence Laboratory, Korea University <i>Graduate Researcher (Advisor: Prof. Daekyum Kim)</i>	March 2025 - Present
Movement Intelligence Laboratory, Korea University <i>Undergraduate Researcher (Advisor: Prof. Daekyum Kim)</i>	November 2023 - February 2025
ROKAF Air Defense Control Command, Osan AB <i>Staff Sergeant (Aviation Control)</i>	October 2020 - July 2022

PUBLICATIONS

IN PREPARATION

[J2] S.H. Kim[†], **S. Baek**[†], B. Hur, Y. Bae, Y. Lee, H.S. Choi, and D. Kim, “Quantitative Ergonomic Assessment of Endoscopists Using Wearable Inertial Measurement Units and Optical Motion Capture”

ACCEPTED

[C1] **S. Baek**[†], Y. Lee[†], B. Hur, E. S. Cha, and D. Kim, “PrintExo: An Open-Source, Shoe-Agnostic, 3D-Printable Ankle Exoskeleton Platform for Accessible Locomotion Research”, in *Proceedings of IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)*, 2026.

PUBLISHED

[J1] B. Hur, **S. Baek**, I. Kang, and D. Kim, “Learning-Based Lower Limb Joint Kinematic Estimation Using Open Source IMU Data”, *Scientific Reports*, vol. 15, no. 1, p. 5287, 2025.

[†] These authors contributed equally.

HONORS & AWARDS

KU Institute for Sustainability BK21 Research Fellowship	2025, 2026
KU Junior Fellow Research Grant	2025
KU Veritas Scholarship (Merit-based Grant)	2023, 2024
Excellence Award, KAP Future Automotive Industry Idea Contest	2024
SW · AI Award(2nd Place), KU AI Forum Poster Presentation	2024

TEACHING

Teaching Assistant, Korea University

MOBI304	<i>Intelligent Mobility System</i>	Spring 2026
MECH485	<i>Introduction to Intelligent Robotics</i>	Fall 2025
MOBI203	<i>Data Structure and Algorithm</i>	Spring 2025
MOBI104	<i>How Things Work</i>	Fall 2024

COMMUNITY

LINC 3.0 KU Entrepreneurial Club, A BOUT

2023

Project Leader

- Led a funded engineering project resulting in a full-scale vessel demonstration on the Han River.
- Conducted data-driven analysis for optimizing river transportation demand and terminal placement.

KU School of Mechanical Engineering, Student Council

2018 - 2019

32nd & 33rd

- Led organizational and managerial efforts for departmental student initiatives and events.

KU Rotaract Club(Community Service & Leadership)

2018 - 2019

External Affairs Executive

- Led and participated in community service activities at local social welfare and disability care centers.